IDEA Student Client – Browser, Console Shortcut and Finding the Servers

Follow-up to the student client image: Koen's feedback of 2026-10-15

Proposal · Status: Draft

IDEA - offline school computing. Prepared for Koen Swings's team.
No real passwords, keys or tokens are included; values in <ANGLE BRACKETS> are placeholders.

Contents

Problem or opportunity 3
1. Browser: Google Chrome (.deb), not Firefox 3
2. One desktop shortcut: "IDEA" opens the Console 4
3. How a client finds the servers 4
  What the server manual sets up today 4
  Options 5
  Recommendation: option C, http://idea.local opens the Console 5
4. Correction: "Manual 2" 6
Decision needed 6
References 6

Author: Steve (Lead Bot) **Date:** 2026-10-15 **Status:** Draft (revised 2026-10-15 after Koen's review: one Console shortcut, no app icons) **Follows:** `2026-10-15-student-client-image.md` (idea#201, approved and merged; frozen)

Problem or opportunity

Koen reviewed the merged student client image and asked for three changes:

1. **Chrome rather than Firefox** as the browser.
2. **An easy way to reach the apps** without typing an address. After review, Koen chose **one desktop shortcut that opens the Console**. No per-app icons: the Console already lists the apps and opens them, and per-app icons would replace that.
3. **No fixed server address.** The merged proposal points every bookmark at `10.42.0.2`. A school can have more than one server Pi, an app can live on any Engine, and an Engine can be switched off or moved. How does a client find the right server?

The merged proposal also says "Manual 2" for the client install manual. That label is confusing. Merged proposals are frozen, so the correction is recorded here (see the last section).

1. Browser: Google Chrome (.deb), not Firefox

Correction. In chat we said Google Chrome does not exist for the Pi. That is no longer true: Google shipped official **arm64 Linux** builds of Chrome on 30 July 2026 (`.deb` for Debian/Ubuntu, `google-chrome-stable_current_arm64.deb`) [1][2]. So Koen can have real Chrome.

Option	Package	Offline / updates	Verdict
Google Chrome stable	Official arm64 `.deb` from Google	Plain `.deb`; we remove Google's apt source and `apt-mark hold` it	**Recommended**
Chromium (Ubuntu)	`chromium-browser` is a **snap** wrapper	Snap refreshes must be held; snap sandbox blocks some features (e.g. app shortcuts) [3]	Not used
Chromium (xtradeb PPA)	Community `.deb` for 26.04 arm64 [4]	Needs pinning so the PPA does not replace other packages; unofficial	**Fallback** if Chrome fails on the Pi
Firefox (merged design)	Mozilla `.deb`	Works	Replaced, per Koen

How it is built into the image (`tools/client-image/customise.sh`):

- Download the pinned Chrome `.deb` once, store it with its SHA-256 in the image build inputs, install with `apt install ./google-chrome-stable_<ver>_arm64.deb`.
- Delete `/etc/apt/sources.list.d/google-chrome.list` and the cron job that re-adds it, then `apt-mark hold google-chrome-stable`. This keeps the decision "updates off; a new tested image is the update".

- Chrome policy file in `/etc/opt/chrome/policies/managed/idea.json`: no first-run screen, no sign-in, no update checks, no default-browser prompt, home page = `http://idea.local` (the Console, section 3).
- Chrome becomes the default browser; Firefox is not installed.
- **Snaps:** none in the student path (unchanged); any snap left in the base stays held.

To test first (risk): Chrome on arm64 is new. Check it starts and plays video on both a **Pi 400 and a Pi 500** with the Ubuntu raspi kernel. If it fails, use the xtradeb Chromium `.deb` with the same hold and policy (`/etc/chromium/policies/managed/`), and tell Koen.

2. One desktop shortcut: "IDEA" opens the Console

The image has **one** shortcut, `idea-console.desktop`, in `/usr/share/applications/` (app menu) and on the learner desktop (GNOME "Desktop Icons" extension, already in Ubuntu Desktop):

```
Name=IDEA
Exec=google-chrome-stable http://idea.local
```

It opens Chrome at `http://idea.local`, which shows the **Console**. The student picks an app there, as today. There are **no per-app icons** and no separate start page: the Console stays the one place that lists and opens apps, so new apps on an Engine appear without a new image. Chrome's home page is the same address.

Nextcloud desktop sync client: not recommended. The computers are shared and students have no personal accounts. A sync client would keep one student's files on the disk for the next student, and would be logged in as whoever used it last. Nextcloud in the browser does not have this problem.

3. How a client finds the servers

What the server manual sets up today

From `docs/manuals/install/install-manual-school-server/`:

- Every school uses the **same addresses**: router `10.42.0.1`, Pi #1 `10.42.0.2`, Pi #2 `10.42.0.3` (DHCP reservations on the router), students on Wi-Fi `10.42.1.0/24`, teachers on `10.42.2.0/24`.
- Each Pi gets a two-word **sticker name** (e.g. `brave-otter`), set as hostname. The Console address on the sticker is `http://<STICKER>.local` (mDNS via avahi).
- Engines already find each other with mDNS and share one Automerge store that lists every Engine, disk and running app. This is what the Console shows.

So two servers per school is already the normal design, and the router reservation only covers two Pis.

Options

Option	Works with many servers / moved apps?	Offline, no per-school config?	Notes
A. Fixed IP `10.42.0.2` in icons	**No.** Breaks if Pi #1 is off or the app is on Pi #2	Yes	Merged design
B. Sticker names `<name>.local`	**No.** The name differs per school, so needs per-school config	No	Fine for humans, not for an image
C. One shared mDNS name `idea.local` answered by **every** Engine, serving the Console	**Yes.** Any live Engine answers; its Console shows all Engines and apps	**Yes**	**Recommended**
D. Static page in the image that queries Engines itself	Yes, but must hard-code IPs or do mDNS in a browser (not possible)	Partly	Logic in the image goes stale; needs a reflash to change

mDNS on clients. Ubuntu Desktop ships avahi and `libnss-mdns`, so `.local` names resolve for every program, and Chrome uses the system resolver for them. mDNS does not cross subnets: the router must reflect mDNS between `10.42.0.0/24` and the Wi-Fi subnets (`avahi` reflector or the OpenWrt `umdns`/`mdns-repeater` package). The Console sticker address `http://<STICKER>.local` already needs this, so it is the same work, done once in `school-router-setup.sh`, the same at every school.

Recommendation: option C, `http://idea.local` opens the Console

- **Every Engine** publishes the extra mDNS name `idea.local`. A request to `http://idea.local` (port 80) serves or redirects to **that Engine's Console**. Nothing new is built for students: the Console already shows every Engine and app from the shared Automerge store, and opens an app on the Engine where it runs.
- No `/go/<app>` links and no extra start page; they would duplicate the Console.
- **Tested on the pool (Axle, 2026-10-15): several Engines cannot all hold `idea.local`.** avahi treats a second claim as a conflict: the first Engine wins and the others fail silently (no `idea-2.local` appears). With one holder it works well: 50/50 lookups answered in about 20 ms, Console HTTP 200 in 10/10 tries. A planned handover (holder stops cleanly) completes in about 6 s.
- **Chosen design: one holder, automatic takeover.** Every Engine runs a small `idea.local` publisher that registers the name through the Pi's own avahi over D-Bus (an address record, `-R`, no second mDNS responder, compatible with `2026-10-13-avahi-hostname-clash.md`). If the name is taken it retries every 2-5 s. The holder serves the Console; when it leaves, another Engine takes the name within seconds. No election via the store.
- **Remaining risk:** a real power-off sends no avahi goodbye, so clients may keep the dead address for up to the 120 s record TTL. This must be proven during implementation by an **automated power-cut test on a pool Pi** (not wizardly-hugle).
- **Backup:** the router's DNS (dnsmasq) also answers `idea.local` with `10.42.0.2` and `10.42.0.3`, set the same way at every school. Either way the shortcut in the image stays the same.

Work:

Who	What
Axle (Engine)	`idea.local` publisher with retry/takeover; port 80 serves/redirects to the Console; automated power-cut test on a pool Pi
Atlas (Ops)	mDNS reflector + dnsmasq fallback in `school-router-setup.sh`; test with 2 Engines, one switched off
Client image worker	Chrome, policies, the one `idea-console.desktop` shortcut
Marco (docs)	the client install manual, the school server install manual, the tutorial

Test: two Engines on one network, from the student Wi-Fi subnet: the shortcut opens the Console. Switch Engine 1 off: the shortcut still opens the Console (on Engine 2). Cut power to the holder: another Engine takes over and clients reach the Console within the 120 s TTL.

4. Correction: "Manual 2"

The merged proposal says "Manual 2" for the client install manual. Use **"the client install manual"** everywhere instead. The folder rename to `install-manual-client` stays a separate PR (as agreed). The implementation branch (`feat/client-image-build`) does not use "Manual 2"; new docs must not either.

Decision needed

1. Google Chrome `.deb` (held, no Google repo) as the browser, xtradeb Chromium as fallback?
2. One desktop shortcut opening Chrome at `http://idea.local`, which shows the Console; no per-app icons (option C, with router mDNS reflector + DNS fallback)?

References

[1] Google, "Download and install Google Chrome" (arm64 .deb for Debian/Ubuntu): https://support.google.com/chrome/answer/95346 [2] Calibre, "Chrome lands on arm64" (30 July 2026): https://calibreapp.com/blog/chrome-for-linux-arm64 [3] Ask Ubuntu, "Installing non-snap version of Chromium": https://askubuntu.com/questions/1569051 [4] XtraDeb Chromium: https://xtradeb.net/apps/chromium/